
X-Ray Data Booklet

Section 1.5 SUBSHELL PHOTOIONIZATION CROSS SECTIONS

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The atomic subshell photoemission cross sections plotted in Fig. 1-4 have been calculated for isolated atoms by Yeh and Lindau [1,2]. The calculations were done with a one-electron central-field frozen-core model using first-order perturbation theory. No single model accurately predicts the photoionization process of all orbitals for all elements from the VUV to 1.5 keV. The complexity of the physics of different atomic orbitals makes it impossible for any single rule to describe all of them. The accuracy of the model used has been discussed in detail by Cooper and Manson [3–5]. A [PDF version of this section](#) is also available.

REFERENCES

1. J.-J. Yeh and I. Lindau, “Atomic Subshell Photoionization Cross Sections and Asymmetry Parameters: $1 < Z < 103$,” *At. Data Nucl. Data Tables* **32**, 1 (1985).
2. J.-J. Yeh, *Atomic Calculations of Photoionization Cross Sections and Asymmetry Parameters* (Gordon and Breach, Langhorne, PA, 1993).
3. J. W. Cooper, *Phys. Rev.* **128**, 681 (1962).
4. S. T. Manson and J. W. Cooper, *Phys. Rev.* **165**, 126 (1968).
5. S. T. Manson, *Adv. Electron. Electron Phys.* **41**, 73 (1976).

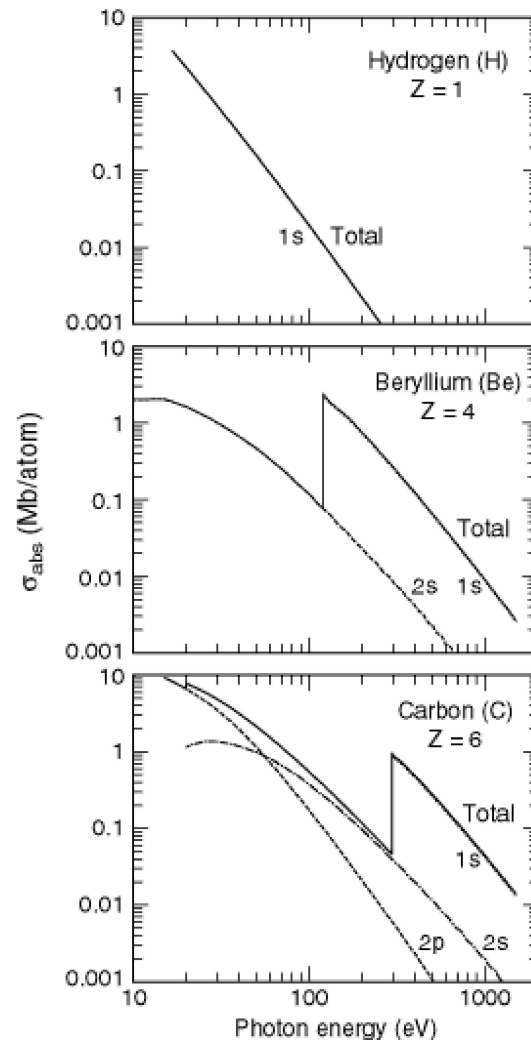


Fig. 1-4. Plots of atomic subshell photoemission cross sections for H, Be and C.

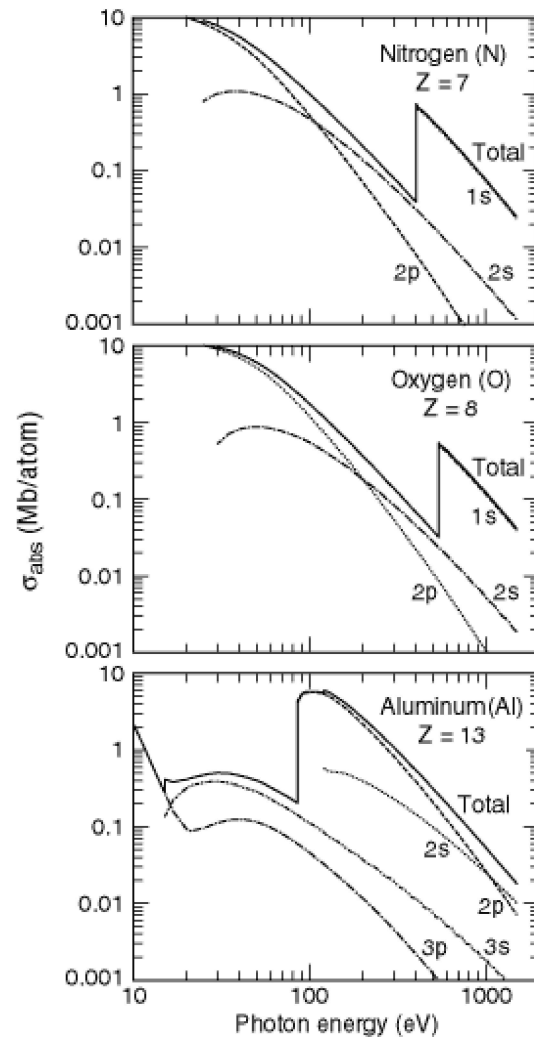


Fig. 1-4. Subshell photoemission cross sections N, O and Al.

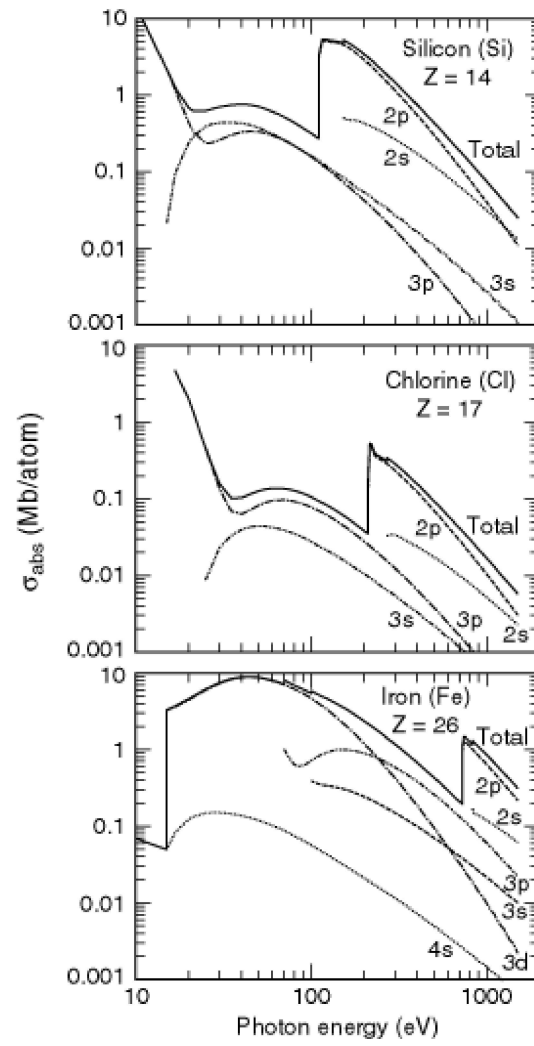


Fig. 1-4. Subshell photoemission cross sections for Si, Cl and Fe.

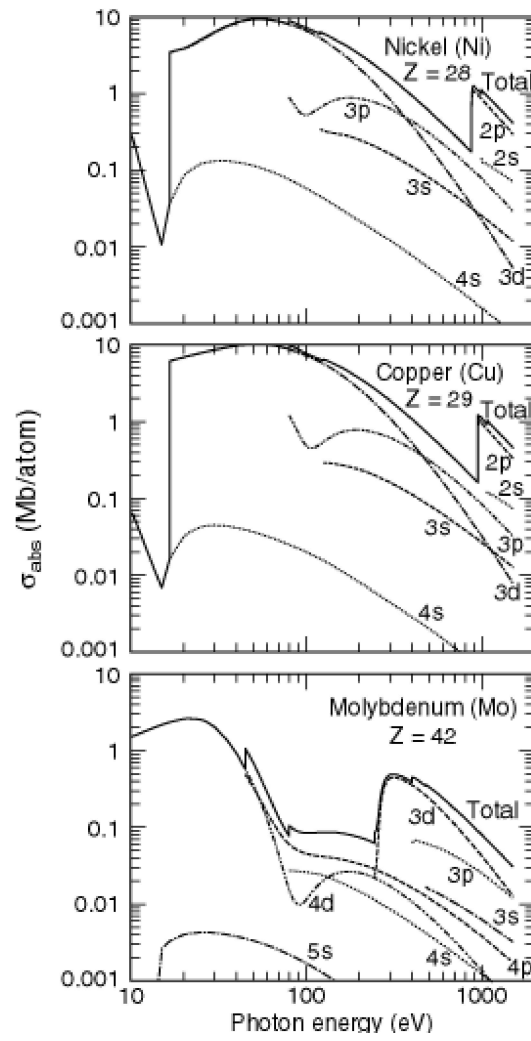


Fig. 1-4. Subshell photoemission cross sections for Ni, Cu and Mo.

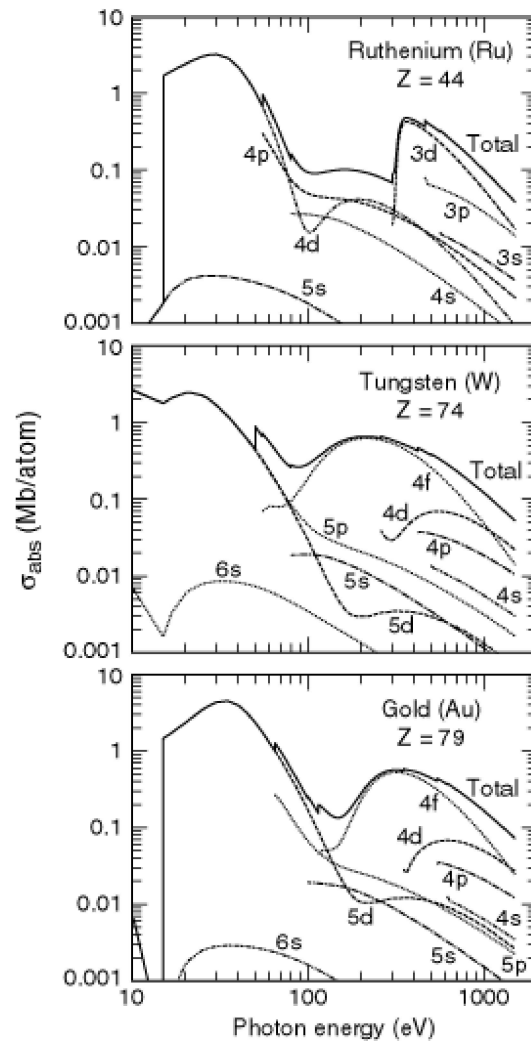


Fig. 1-4. Subshell photoemission cross sections for Ru, W and Au.